

# WASP-163b

R-2340

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-163\_210611 · created 2026-08-19T00:18:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                              |
|---------------------------------------------------|------------------------------|
| Mid-Transit Time (Tmid)                           | 2459376.833 +/- 0.02 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.146 +/- 0.038              |
| Transit depth (Rp/Rs)^2                           | 2.13 +/- 1.12 [%]            |
| Orbital Inclination (inc)                         | 83.18 +/- 2.65               |
| Airmass coefficient 1 (a1)                        | 1.009 +/- 0.017              |
| Airmass coefficient 2 (a2)                        | -0.007 +/- 0.029             |
| Scatter in the residuals of the lightcurve fit is | 1.72 %                       |
| Best Comparison Star                              | #1 - [542, 41]               |
| Optimal Method                                    | PSF photometry               |
| Transit Duration (day)                            | 0.078 +/- 0.023              |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R* fit vs published    | 0.146 ± 0.038 vs 0.119 (deviation 0.71σ) |
| Duration fit vs published | 0.078 d vs 0.0942 d (deviation 0.7σ)     |
| Mid-time O-C              | -10.7 min                                |
| Depth SNR                 | 3.8                                      |
| Residual scatter          | 1.72 %                                   |

## Light curve

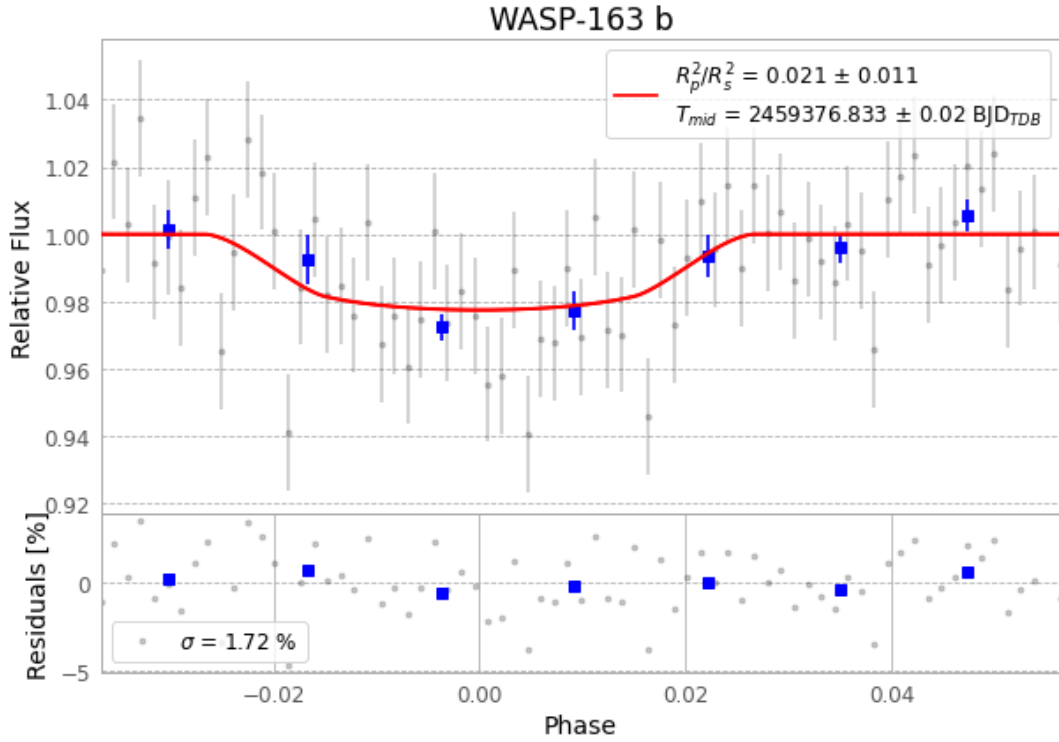


Figure 1 — FinalLightCurve\_WASP-163 b\_2021-06-10.png (SHA-256 c73cc4d0c0733926...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 48], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 165af04f8c1c3a82...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

76 FITS frames (50 MB), WASP-163210611062412.FITS → WASP-163210611100912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-210611.log (1ef158bf9752...), FinalLightCurve\_WASP-163 b\_2021-06-10.png (c73cc4d0c073...), FinalLightCurve\_WASP-163 b\_2021-06-10.csv (c9bcd7a9a14b...)

## Compute resources

Wall time 123.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 00:18Z.